Year 10 Foundation

REVISION PACK

Number 3

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Number 3 · Laws of indices (Lesson 6.1)

Q	Working	Answer
writing as a single power		
1.1	**(a)** 4^{5+2} **(b)** 8^{10-3} **(c)** $5^{2\times 3}$	**(a)** 4^{7} **(b)** 8^{7} **(c)** 5^{6}
1.2	**(a)** 6^{2+7} **(b)** 9^{8-5} **(c)** $7^{4\times 2}$	**(a)** 6^{9} **(b)** 9^{3} **(c)** 7^{8}
1.3	**(a)** 3^{4+6} **(b)** 2^{9-2} **(c)** $2^{5\times 3}$	**(a)** 3^{10} **(b)** 2^{7} **(c)** 2^{15}
1.4	**(a)** 10^{3+5} **(b)** 5^{12-4} **(c)** $4^{3\times 4}$	**(a)** 10^{8} **(b)** 5^{8} **(c)** 4^{12}
1.5	**(a)** 7^{2+8} **(b)** 3^{11-6} **(c)** $6^{2\times 6}$	**(a)** 7^{10} **(b)** 3^{5} **(c)** 6^{12}
1.6	**(a)** 2^{6+5} **(b)** 6^{9-3} **(c)** $9^{3\times 3}$	**(a)** 2^{11} **(b)** 6^{6} **(c)** 9^{9}
negative and zero powers		
1.7	**(a)** $4^{-2} = \frac{1}{4^{2}}$ **(b)** any number to the power 0 is 1	**(a)** $\frac{1}{16}$ **(b)** **1**

1.8	**(a)** $2^{-4} = \frac{1}{2^4}$ **(b)** any number to the power 0 is 1	**(a)** $\frac{1}{16}$ **(b) 1**
1.9	**(a)** $10^{-2} = \frac{1}{10^2}$ **(b)** any number to the power 0 is 1	**(a)** $\frac{1}{100}$ **(b) 1**
1.10	**(a)** $3^{-3} = \frac{1}{3^3}$ **(b)** any number to the power 0 is 1	**(a)** $\frac{1}{27}$ **(b) 1**
1.11	**(a)** $5^{-2} = \frac{1}{5^2}$ **(b)** any number to the power 0 is 1	**(a)** $\frac{1}{25}$ **(b) 1**
1.12	**(a)** $6^{-1} = \frac{1}{6^1}$ **(b)** any number to the power 0 is 1	**(a)** $\frac{1}{6}$ **(b) 1**

Number 3 · Using index laws (Lesson 6.1)

Q	Working	Answer
working out with powers		
2.1	**(a)** 2^{4+3} **(b)** $5^{3-1} = 5^2$ $= 25$ **(c)** $9^0 = 1$	**(a)** 2^7 **(b) 25** **(c) 1**
2.2	**(a)** 4^{2+5} **(b)** $3^{4-1} = 3^3$ $= 27$ **(c)** $12^0 = 1$	**(a)** 4^7 **(b) 27** **(c) 1**

2.3	**(a)** 6^{3+2} **(b)** $2^{5-1} = 2^4$ $= 16$ **(c)** $50^0 = 1$	**(a)** 6^5 **(b) 16** **(c) 1**
2.4	**(a)** 7^{5+2} **(b)** $10^{4-1} = 10^3$ $= 1000$ **(c)** $3^0 = 1$	**(a)** 7^7 **(b) 1000** **(c) 1**
2.5	**(a)** 5^{2+6} **(b)** $6^{3-1} = 6^2$ $= 36$ **(c)** $8^0 = 1$	**(a)** 5^8 **(b) 36** **(c) 1**
2.6	**(a)** 3^{5+4} **(b)** $2^{6-1} = 2^5$ $= 32$ **(c)** $11^0 = 1$	**(a)** 3^9 **(b) 32** **(c) 1**

Number 3 · Finding the power (Lesson 6.1)

Q	Working	Answer
finding the power n		
3.1	$32 = 2^5$	$n = 5$
3.2	$25 = 5^2$, so $\frac{1}{25} = 5^{-2}$	$n = -2$
3.3	$8 = 2^3$, so $\frac{1}{8} = 2^{-3}$	$n = -3$
3.4	$0.001 = \frac{1}{1000} = 10^{-3}$	$n = -3$
3.5	Any number to the power 0 is 1	$n = 0$
3.6	$81 = 3^4$, so $\frac{1}{81} = 3^{-4}$	$n = -4$